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United States Patent	12401070
Kind Code	B2
Date of Patent	August 26, 2025
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BMS wakeup device and method

Abstract

A battery pack according to an embodiment of the present invention includes a (+) output terminal of the battery pack, a (-) output terminal of the battery pack, a (-) line connecting the (-) output terminal to a (-) terminal of a battery cell constituting the battery pack, a pre-charge FET provided on the (-) line, a system contact terminal configured to check whether the external system is connected, a battery management system (BMS), and a charge/discharge FET configured to charge or discharge the battery pack under a control of the BMS, wherein the pre-charge FET and the charge/discharge FET are connected in parallel.

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Appl. No.:	17/633404
Filed (or PCT Filed):	October 30, 2020
PCT No.:	PCT/KR2020/014978
PCT Pub. No.:	WO2021/096116
PCT Pub. Date:	May 20, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220294034 A1	Sep. 15, 2022

Foreign Application Priority Data

KR

10-2019-0146382

Nov. 15, 2019

Publication Classification

Int. Cl.: H01M10/42 (20060101); H01M10/44 (20060101); H01M10/46 (20060101);
H01M10/48 (20060101); H02J7/00 (20060101)

U.S. Cl.:

CPC H01M10/4257 (20130101); H01M10/443 (20130101); H01M10/46 (20130101);
H01M10/486 (20130101); H02J7/0063 (20130101); H01M2010/4271 (20130101)

Field of Classification Search

CPC: H01M (10/4257); H01M (10/443); H01M (10/46); H01M (10/486); H01M (10/48);
H01M (10/44); H01M (2010/4271); H01M (2220/30); H02J (7/0063); H02J (7/00304);
H02J (7/0036); H02J (7/0047); H02J (7/0068); H02J (7/00); Y02E (60/10)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5477130	12/1994	Hashimoto et al.	N/A	N/A
2007/0257635	12/2006	Yang	N/A	N/A
2009/0087722	12/2008	Sakabe et al.	N/A	N/A
2009/0111005	12/2008	Hur	N/A	N/A
2012/0286575	12/2011	Park et al.	N/A	N/A
2013/0234508	12/2012	Eisele et al.	N/A	N/A
2015/0219720	12/2014	Huh	N/A	N/A
2015/0222117	12/2014	Im	N/A	N/A
2018/0015833	12/2017	Namou et al.	N/A	N/A
2018/0115178	12/2017	Moon et al.	N/A	N/A
2019/0067957	12/2018	Yeom	N/A	N/A
2019/0280341	12/2018	Li	N/A	N/A
2020/0052346	12/2019	Park	N/A	N/A
2021/0143485	12/2020	Kim	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
101083402	12/2006	CN	N/A
101425693	12/2008	CN	N/A
103069641	12/2012	CN	N/A
208782521	12/2018	CN	N/A
110226258	12/2018	CN	N/A
1 855 368	12/2006	EP	N/A
2000-060006	12/1999	JP	N/A
2009-089487	12/2008	JP	N/A

2009-112192	12/2008	JP	N/A
2010-193558	12/2009	JP	N/A
2013188088	12/2012	JP	B60L 3/00
2013188088	12/2012	JP	N/A
2014-045551	12/2013	JP	N/A
5768780	12/2014	JP	N/A
2015-173569	12/2014	JP	N/A
10-20140025627	12/2013	KR	N/A
10-20150091890	12/2014	KR	N/A
10-1741643	12/2016	KR	N/A
10-20170060406	12/2016	KR	N/A
101741643	12/2016	KR	H01M 10/425
10-20180045954	12/2017	KR	N/A
10-20180053843	12/2017	KR	N/A
10-2018-0093322	12/2017	KR	N/A
10-20180087013	12/2017	KR	N/A
10-20190051477	12/2018	KR	N/A
2019/078589	12/2018	WO	N/A

OTHER PUBLICATIONS

English translation of JP2013188088. cited by examiner

English translation of KR101741643. cited by examiner

International Search Report and Written Opinion issued in corresponding International Patent Application No. PCTKR2020/014978, dated Oct. 30, 2020. cited by applicant

Office Action issued Jun. 17, 2023, in corresponding Chinese Patent Application No. 202080020236.5. cited by applicant

Extended European Search Report dated May 25, 2022 issued by the European Patent Office in corresponding European Patent Application No. 20886278.9. cited by applicant

Office Action dated Oct. 11, 2022, issued by the Japanese Patent Office in corresponding Japanese Patent Application No. 2021-556598. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a device and method for waking up a BMS using the power of a battery itself without applying power from the outside.

BACKGROUND ART

(2) In recent years, as the demand for portable electronic products such as laptops, video cameras, and portable phones is rapidly increasing, and the development of energy storage batteries, robots, satellites, and the like begins in full swing, research on a high-performance secondary battery capable of repetitive charging and discharging is actively being conducted.

(3) Currently commercialized secondary batteries include nickel cadmium batteries, nickel hydride batteries, nickel zinc batteries, and lithium secondary batteries, and among them, lithium secondary batteries are in the spotlight due to their advantages such as free charging and discharging, a very

low self-discharge rate, and high energy density, as a memory effect rarely occurs compared to nickel-based secondary batteries.

(4) Battery packs are used in various fields, and often require large capacity, such as electric powered vehicles or smart grid systems. In order to increase the capacity of the battery pack, there may be a method of increasing the capacity of the secondary battery, that is, the battery cell itself but in this case, the effect of increasing the capacity is not large, and there is a disadvantage in that there is a physical limitation on the size expansion of the secondary battery. Therefore, in general, a battery pack in which a plurality of battery modules are connected in series and in parallel is widely used.

(5) These battery packs often include a battery management system (BMS) that manages battery modules. Furthermore, the BMS monitors the temperature, voltage, and current of the battery module, and controls the balancing operation, cooling operation, charging operation or discharging operation of the battery pack based on the monitored battery module state.

(6) In particular, a large number of battery packs, such as the BMS provided in a 12V lithium battery pack, can operate by receiving power from a battery module included in the battery pack. However, in the case of such a configuration, even if the value is small due to the current consumption of the BMS, there is a problem of consuming power of the battery pack.

(7) In order to solve this problem, a technique has been proposed in which the BMS is not always turned on and is turned on only when its operation is required. However, in this case, when the operation of the BMS is required, for example, a technique for waking up the BMS when connected to an external device is required.

(8) But, in order to apply separate power from the outside, the conventional circuit implementing the BMS wake-up technology has a disadvantage in that complex circuits such as 12V Aux-related circuits, connectors, and charge pumps are used.

(9) Therefore, in the present invention, when the battery is connected to an external device to solve this disadvantage, suggested is a device and method for waking up the BMS with only the power source of the battery. (Patent Document 0001) Korean Laid-Open Patent Publication No. 10-2019-0051477

DISCLOSURE

Technical Problem

(10) The present invention provides a device and method for waking up a BMS only with a power source from a battery when the battery is connected to an external device.

Technical Solution

(11) A battery pack according to an embodiment of the present invention includes a (+) output terminal of the battery pack, a (-) output terminal of the battery pack, a (-) line connecting the (-) output terminal to a (-) terminal of a battery cell constituting the battery pack, a pre-charge FET provided on the (-) line, a system contact terminal configured to check whether the external system is connected, a battery management system (BMS), and a charge/discharge FET configured to charge or discharge the battery pack under a control of the BMS, wherein the pre-charge FET and the charge/discharge FET are connected in parallel.

(12) When the battery pack and the external device are connected, the (+) output terminal of the battery pack and the system contact terminal of the battery pack may be electrically connected due to a short member provided in the external device.

(13) When the (+) output terminal and the system contact terminal are connected (shorted) through the short member, the pre-charge FET may be turned on, wherein when the pre-charge FET is turned on, a voltage of the battery pack is supplied to the BMS through the (+) output terminal and the pre-charge FET, so that the BMS may wake up.

(14) After the BMS wakes up, the BMS turns off the pre-charge FET.

(15) After the BMS wakes up, the BMS may check voltage, current, and temperature of the battery, and turn on the charge/discharge FET when all of the voltage, current, and temperature are not

abnormal.

(16) The external device may be any one of a mobile phone, a tablet computer, a notebook computer, a power tool, a wearable electronic device, and a power storage device, in which the battery pack is mounted.

(17) A method of waking up a battery management system (BMS) of a battery pack according to an embodiment of the present invention includes an external device connection step in which the battery pack is connected to an external device, a system contact terminal connection step in which a (+) terminal of the battery pack and a system contact terminal of the battery pack are connected (shorted) while the battery pack is connected to the external device, a pre-charge FET on step in which the (+) terminal of the battery pack and the system contact terminal are connected in the system contact terminal connection step to turn on a pre-charge FET, and a BMS wake-up step in which when the pre-charge FET of the battery pack is turned on, an output of the battery pack is supplied to the BMS through the pre-charge FET at the (+) terminal to wake up the BMS.

(18) In the system contact terminal connection step, the (+) terminal of the battery pack and the system contact terminal of the battery pack may be connected by a short member provided in the external device.

(19) After the BMS wakes up in the BMS wake-up step, the BMS may perform a pre-charge FET off step of turning off the pre-charge FET.

(20) After the BMS wakes up in the BMS wake-up step, the BMS may perform a battery state measurement step of measuring voltage, current, and temperature of the battery pack, so that when the voltage, current, and temperature of the battery pack measured in the battery state measurement step are within a normal range, the charge/discharge FET on step of turning on the charge/discharge FET is performed.

Advantageous Effects

(21) In the present invention, when the battery is connected to an external device, a separate wake-up circuit is not required because the BMS can be wake-up only with the power source of the battery.

(22) In addition, since the present invention does not require a separate wake-up circuit, the utility of the internal space of the battery can be improved.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a view showing a battery pack according to an embodiment of the present invention.

(2) FIG. 2 is a view showing a battery pack according to another embodiment of the present invention.

(3) FIG. 3 is a flowchart illustrating a BMS wake-up method of a battery pack according to an embodiment of the present invention.

MODE FOR INVENTION

(4) Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings so that those of ordinary skill in the art may easily implement the present invention. However, the present invention may be implemented in various forms and is not limited to the embodiments described herein. In the drawings, parts irrelevant to the description are omitted in order to clearly describe the present invention, and like reference numerals refer to like elements throughout the specification.

(5) Terms including ordinal numbers, such as first and second, may be used to describe various elements, but the elements are not limited by the terms. The above terms are used only to distinguish one component from another. For example, a first component may be referred to as a second component and vice versa without departing from the scope of the present invention. The

terms used in the present application are used only to describe specific embodiments, and are not intended to limit the present invention. The terms of a singular form may include plural forms unless otherwise specified.

(6) Throughout the specification, when one part is “connected” to another part, this includes not only “directly connected” cases, but also “electrically connected” cases with other elements therebetween. In addition, when a certain part “includes” a certain component, it means that other components may be further included rather than excluding other components unless otherwise stated. As used throughout the present specification, the term “-ing step” or “a step of -ing” does not mean “a step for -ing”.

(7) As for terms used in the present invention, general terms that are currently widely used as possible are selected while considering functions in the present invention, but this may vary depending on the intentions or precedents of those skilled in the art, and the emergence of new technologies. In addition, in certain cases, there are terms arbitrarily selected by the applicant, and in this case, the meaning of the terms will be described in detail in the description of the corresponding invention. Therefore, the terms used in the present invention should be defined based on the meaning of the term and the overall contents of the present invention, not a simple name of the term.

1. Battery Pack According to Embodiment of Present Invention

(8) FIG. 1 is a view showing a battery pack according to an embodiment of the present invention.

(9) Hereinafter, a battery pack according to an embodiment of the present invention will be described with reference to FIG. 1.

(10) A battery pack according to an embodiment of the present invention may be configured to include a (+) output terminal **10** of the battery pack, a (-) output terminal **20** of the battery pack, a (-) line **2** connecting the (-) output terminal **20** and the (-) terminal of a battery cell constituting the battery pack, a pre-charge FET **210** provided on the (-) line **2**, a system contact terminal **30** that checks whether an external system is connected, a Battery Management System (BMS) **100** and, a charge/discharge FET **220** that charges or discharges the battery pack under the control of the BMS **100**.

(11) Specifically, in relation to a battery pack according to an embodiment of the present invention, when the battery pack and external device are connected, due to a short member **300** provided in the external device, the (+) output terminal **10** of the battery pack and the system contact terminal **30** of the battery pack are electrically connected (shorted) so that the BMS **100** may wake up.

(12) For example, the short member **300** provided in the external device is formed in a rod shape of a conductive material and in addition, one end of the short member **300** may contact the (+) output terminal **10** of the battery pack, and the other end may contact the system contact terminal **30** of the battery pack. Meanwhile, the rest of the short member **300** except for one end and the other end may be wrapped with an insulating member.

(13) On the other hand, in relation to the battery pack according to an embodiment of the present invention, when the (+) output terminal **10** of the battery pack and the system contact terminal **30** are connected (shorted) through the short member **300** provided in the external device, the pre-charge FET **210** may be turned on.

(14) And, when the pre-charge FET **210** is turned on, the voltage of the battery pack is supplied to the BMS **100** through the (+) output terminal **10** and the pre-charge FET **210** so that the BMS **100** may wake up.

(15) In other words, when the (+) output terminal **10** of the battery pack and the system contact terminal **30** are connected (shorted) through the short member **300**, the power path connected in the (+) output terminal **10** of the battery pack->the short member **300**->the system contact terminal **30**->the pre-charge FET **210**->the BMS **100** is formed so that it is possible to wake up the BMS **100** without a separate external power source.

(16) Meanwhile, the external device may be any one of a mobile phone, a tablet computer, a

notebook computer, a power tool, a wearable electronic device, and a power storage device, in which the battery pack is mounted.

(17) Meanwhile, the BMS **100** may turn off the pre-charge FET **210** after the BMS **100** wakes up.
(18) And, after the BMS **100** wakes up, the BMS **100** may check the voltage, current, and temperature of the battery, and when there is no abnormality in all of the voltage, current, and temperature, turn on the charge/discharge FET **220**.

(19) Accordingly, the battery pack may wake up without a separate power supply from the outside, and normally perform charging/discharging.

2. Battery Pack According to Another Embodiment of Present Invention

(20) FIG. **2** is a view showing a battery pack according to another embodiment of the present invention.

(21) Hereinafter, a battery pack according to another embodiment of the present invention will be described with reference to FIG. **2**.

(22) A battery pack according to another embodiment of the present invention includes all the configurations of the battery pack of the present invention described above, and includes a short member inside the battery pack.

(23) Specifically, a battery pack according to an embodiment of the present invention may be configured to include a (+) output terminal **10** of the battery pack, a (-) output terminal **20** of the battery pack, a (-) line **2** connecting the (-) output terminal **20** and the (-) terminal of a battery cell constituting the battery pack, a pre-charge FET **210** provided on the (-) line **2**, a system contact terminal **30** that checks whether an external system is connected, a Battery Management System (BMS) **100**, a charge/discharge FET **220** that charges or discharges the battery pack under the control of the BMS **100**, and a short part.

(24) On the other hand, in the battery pack according to another embodiment of the present invention, other configurations and operations except for the short part are the same as those of the battery pack according to the embodiment of the present invention, so detailed descriptions are omitted.

(25) Hereinafter, only the structure and operation of the short member of the battery pack according to another embodiment of the present invention will be described.

(26) The short part according to another embodiment of the present invention may include a first support **410** located next to the (+) output terminal of the battery pack, a second support **420** located next to the system contact terminal, a short bar **430** supported by the first and second supports **410** and **420**, and made of a conductive material.

(27) The first and second supports **410** and **420** are formed of a material that can be contracted/expanded by an external pressing pressure, and when pressing pressure is applied from the outside, one end of the short bar **430** may contact the (+) output terminal **10** of the battery pack, and the other end of the short bar **430** may contact the system contact terminal **30**.

(28) For example, the first support **410** may be composed of a first spring, and one end of the first spring may be fixed around the battery pack case in which the (+) output terminal of the battery pack is formed, and the other end may be connected to one end of the short bar, and the second support **420** may be composed of a second spring, and one end of the second spring may be fixed around a battery pack case in which the system contact terminal is formed, and the other end may be connected to the other end of the short bar.

(29) Meanwhile, in relation to the short bar, when a battery is connected to an external device and a pressing pressure is applied to the short bar **430** by the external device, the first and second members contract, and as a result, the short bar **430** contacts the (+) output terminal **10** and the system contact terminal **30** of the battery pack.

(30) In other words, when the (+) output terminal **10** of the battery pack and the system contact terminal **30** are connected (shorted) through the short bar **430**, the power path connected in the (+) output terminal **10** of the battery pack->the short bar **430**->the system contact terminal **30**->the pre-

charge FET **210**->the BMS **100** is formed so that it is possible to wake up the BMS **100** without a separate external power source.

(31) Meanwhile, the short bar **430** may be wrapped with an insulating member except for a portion corresponding to the (+) output terminal and a portion corresponding to the system contact terminal of the battery pack.

3. BMS Wake-Up Method of Battery Pack According to Embodiment of Present Invention

(32) FIG. **3** is a flowchart illustrating a BMS wake-up method of a battery pack according to an embodiment of the present invention.

(33) Hereinafter, a BMS wake-up method of a battery pack according to an embodiment of the present invention will be described with reference to FIG. **3**.

(34) The BMS wake-up method of a battery pack according to an embodiment of the present invention may be configured to include an external device connection step **S100** in which the battery pack is connected to an external device, a system contact terminal connection step **S200** in which the (+) terminal of the battery pack and the system contact terminal of the battery pack are connected (shorted) while the battery pack is connected to an external device, a pre-charge FET on step **S300** in which the (+) terminal of the battery pack and the system contact terminal are connected to turn on the pre-charge FET in the system contact terminal connection step, and a BMS wake-up step **S400** in which when the pre-charge FET of the battery pack is turned on, the output of the battery pack is supplied to the BMS through the pre-charge FET at the (+) terminal.

(35) Specifically, the system contact terminal connection step **S200** is a step in which the (+) terminal of the battery pack and the system contact terminal of the battery pack are connected by a short member.

(36) For example, the short member is formed in a rod shape of a conductive material and in addition, one end of the short member may contact the (+) output terminal of the battery pack, and the other end may contact the system contact terminal of the battery pack. In addition, the rest of the part except for the one end and the other end may be wrapped with an insulating member.

(37) In addition, when the (+) output terminal of the battery pack and the system contact terminal are connected (shorted) through the short member, the pre-charge FET on step may be performed.

(38) And when the pre-charge FET on step **S300** is performed, a BMS wake-up step in which a voltage of a battery pack is supplied to the BMS through the (+) output terminal and the pre-charge FET may be performed to wake up the BMS.

(39) In other words, when the (+) output terminal of the battery pack and the system contact terminal are connected (shorted) through the short member, a power path connected in the battery pack's (+) output terminal->the system contact terminal->the pre-charge FET->the BMS is formed, so that the BMS can wake up.

(40) Meanwhile, after the BMS wakes up in the BMS wake-up step **S400**, the BMS may perform a pre-charge FET off step of turning off the pre-charge FET.

(41) In addition, after the BMS wakes up in the BMS wake-up step **S400**, the BMS may perform a battery state measurement step of measuring voltage, current, and temperature of the battery pack, and when the voltage, current, and temperature of the battery pack measured in the battery state measurement step are within the normal range, perform a charge/discharge FET on step of turning on the charge/discharge FET.

(42) On the other hand, although the technical idea of the present invention has been specifically described according to the above embodiment, it should be noted that the above embodiments are for the purpose of explanation and not limitation. In addition, those skilled in the art in the technical field of the present invention will be able to understand that various embodiments are possible within the scope of the spirit of the present invention.

Claims

1. A battery pack in which a battery management system (BMS) is configured to wake up after being connected to an external system, the battery pack comprising: a (+) output terminal of the battery pack; a (−) output terminal of the battery pack; a (−) line connecting the (−) output terminal to a (−) terminal of a battery cell constituting the battery pack; a pre-charge field-effect transistor (FET) provided on the (−) line; a system contact terminal configured to check whether the external system is connected; the BMS; and a charge/discharge FET configured to charge or discharge the battery pack under a control of the BMS, wherein the pre-charge FET and the charge/discharge FET are connected in parallel, wherein, when the battery pack and the external device are connected, the (+) output terminal of the battery pack and the system contact terminal of the battery pack are electrically connected due to a short member provided in the external device, and wherein: when the (+) output terminal and the system contact terminal are connected and shorted through the short member, the pre-charge FET is turned on; and when the pre-charge FET is turned on, a voltage of the battery pack is supplied to the BMS through the (+) output terminal and the pre-charge FET, so that the BMS wakes up.
 2. The battery pack of claim 1, wherein, after the BMS wakes up, the BMS is further configured to turn off the pre-charge FET.
 3. The battery pack of claim 1, wherein, after the BMS wakes up, the BMS is further configured to: check voltage, current, and temperature of the battery, and turn on the charge/discharge FET when all of the voltage, current, and temperature are not abnormal.
 4. The battery pack of claim 1, wherein the external device includes any one of: a mobile phone, a tablet computer, a notebook computer, a power tool, a wearable electronic device, and a power storage device, in which the battery pack is mounted.
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